
< *Self-Care Website* >

B.TECH. MINOR PROJECT REPORT

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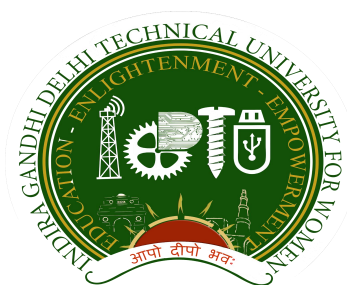
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STUDENT(s) UNDERTAKING

This is to undertake that the work titled “**Self-Care Website**” in the B.Tech. Minor Project report during academic year 2024-25 under the guidance of **Dr. Himanshu Mittal** is our original work.

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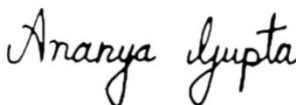
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CERTIFICATE

This is to certify that the project work titled “**Self-Care Website**” submitted by **Ridhima Mishra (015), Deeksha Sharma (040) and Ananya Gupta (075)** in this B.Tech. Minor Project Report during the academic year 2024-25 was done under my guidance and supervision.

This work is original work to the best of my knowledge and has not been submitted anywhere else for the award of any credits / degree whatsoever. The work is satisfactory for the award of B.Tech. Minor Project credits.

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ACKNOWLEDGEMENT

We are sincerely grateful to our supervisor, **Dr. Himanshu Mittal**, at the Department of Artificial Intelligence and Data Science, for his invaluable guidance, insightful suggestions, and support throughout the development of our minor project, “**Self-Care Website**”. His encouragement motivated us to explore innovative solutions and successfully complete this project.

We extend our heartfelt thanks to the **Department of Artificial Intelligence and Data Sciences, Indira Gandhi Delhi Technical University for Women**, for providing us with a conducive and resourceful environment to work on our project. Their unwavering support and encouragement allowed us the freedom to explore creative approaches and tackle a complex problem effectively.

Finally, we appreciate the collaborative spirit and the opportunities afforded to us, which made this endeavor a meaningful and enriching experience.

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ABSTRACT

Self-Care Website

In today's fast-paced and digitally driven world, individuals increasingly struggle to maintain their mental and emotional well-being, often feeling overwhelmed by stress, anxiety, and burnout. Although countless self-help resources are available, navigating these options and finding the right tools tailored to individual needs can be confusing and time-consuming. Many lack the guidance and structure to establish sustainable self-care habits, leading to further frustration. There is a growing demand for an all-in-one solution that simplifies self-care by offering personalized self-help book recommendations, intuitive habit-tracking tools, and an accessible therapy chatbot. By providing these features in a cohesive platform, users can receive tailored support, stay accountable to their wellness goals, and foster a healthier, more balanced lifestyle.

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1. INTRODUCTION

Project Overview:

A comprehensive web platform offering virtual counseling to support users' mental, physical, and emotional wellbeing.

Key Features:

➤ **Mental Health Support:**

- ✧ Provides a virtual counselor for empathetic, personalized conversations.
- ✧ Recommends self-care routines that users can track.
- ✧ Powered by a dataset featuring real conversations and expert advice given by professional counsellors.

➤ **Habit Tracker:**

- ✧ Allows users to set goals and monitor progress.
- ✧ User can add the recommendations from the virtual counselor to foster personal growth.

➤ **Book Recommendation System:**

- ✧ Suggests relevant books and journals.
- ✧ Tailored to each user's mental health, self-improvement, and development goals.

Objective:

Empower users to take charge of their mental health and lifestyle improvements through personalized, accessible resources.

BACKGROUND

In today's fast-paced world, maintaining mental, physical, and emotional well-being has become more challenging than ever. The increasing demands of modern life, coupled with the impacts of events like the COVID-19 pandemic, have significantly strained individuals' mental health. Reports from the World Health Organization (WHO) reveal that mental health disorders such as anxiety, depression, and burnout are among the leading causes of disability worldwide, with millions of people affected.

While there is growing awareness of the importance of mental health, access to professional care remains limited for many individuals. Key barriers include:

- 1. Financial Constraints:** Professional counseling and therapy services can be expensive, making them inaccessible for people with limited financial resources.
- 2. Geographical Limitations:** In rural or remote areas, access to licensed mental health professionals is often scarce.
- 3. Cultural Stigma:** In many societies, mental health issues are stigmatized, discouraging individuals from seeking help.

In response to these challenges, digital solutions have emerged to provide alternative means of support. Self-help websites, mindfulness apps, and habit-tracking platforms are some examples. However, most of these tools focus on singular aspects of well-being and fail to address the interplay between mental, physical, and emotional health. Many existing platforms lack personalization and meaningful engagement, leading to poor user retention and limited long-term benefits.

There is a clear need for a more integrated approach—a platform that brings together mental health support, lifestyle habit tracking, and personalized self-improvement resources to help individuals take charge of their overall well-being.

PROBLEM STATEMENT

Despite the advancements in technology and the availability of various digital tools, there are several shortcomings in the existing solutions:

1. Lack of Personalization: Current mental health tools often provide generalized advice that does not cater to individual preferences, life circumstances, or specific mental health conditions. This lack of empathy and relevance makes the solutions less effective.

2. Fragmented Experience: Users are often forced to switch between different platforms—one for mental health support, another for habit tracking, and yet another for accessing self-help resources like books or journals. This fragmentation leads to inefficiency and reduced motivation.

3. Limited Accessibility: Professional mental health services remain out of reach for many due to high costs or the unavailability of licensed therapists in certain regions. While some free digital tools exist, they often lack depth and reliability.

4. Underutilization of Technology: Although technologies like natural language processing (NLP) and machine learning have advanced significantly, many platforms fail to leverage these capabilities to provide intelligent and engaging user interactions.

These challenges emphasize the need for a comprehensive, integrated platform that offers personalized, accessible, and empathetic support across multiple dimensions of well-being.

MOTIVATION

The "**Self-Care Website**" project is motivated by the belief that everyone deserves accessible, high-quality tools to support their mental, physical, and emotional health. The key drivers of this project are:

1. Addressing the Mental Health Crisis: With mental health conditions on the rise globally, this project aims to provide a scalable solution that can support individuals in managing their

mental health effectively. The integration of a virtual counselor, built using Python, ensures that users receive personalized and empathetic responses, making mental health support more approachable.

2. Promoting Holistic Well-being: Mental health, physical health, and emotional stability are interconnected. By combining features like habit tracking and book recommendations with mental health support, the platform promotes a well-rounded approach to self-care.

3. Empowering Users Through Technology: The project leverages technologies such as Python, Flask, and data scraping to provide a seamless and engaging experience. For instance, the virtual counselor is trained using data scraped from conversations on counselchat.com, ensuring it offers advice similar to that provided by professional therapists.

4. Bridging Accessibility Gaps: By offering a free or low-cost solution, the project reduces barriers to accessing mental health resources, especially for individuals who cannot afford professional services or face stigma in seeking help.

5. Fostering Self-Improvement: The habit tracker and book recommendation system empower users to set goals, monitor their progress, and access resources that align with their personal development journey. These features encourage users to take proactive steps toward improving their lives.

6. Innovating in the Self-Care Domain: The project fills a significant gap in the market by integrating multiple well-being tools into a single platform. Unlike fragmented solutions, this website offers a cohesive experience that addresses various aspects of a user's life.

By combining mental health support, habit tracking, and book recommendations, the "Self-Care Website" serves as a holistic tool for individuals to take control of their well-being. The project's emphasis on personalization, accessibility, and ease of use ensures that it has the potential to make a meaningful impact on users' lives.

2. RELATED WORK

The self-care and mental health domain has seen several advancements in tools designed to offer counseling, habit tracking, and personal development assistance. However, most existing platforms focus narrowly on specific aspects and lack a holistic approach. Below is a detailed review of four platforms that aim to provide counseling or mental health support, emphasizing their strengths and limitations.

1. “Woebot”

Woebot is an AI-driven chatbot designed to support mental health by using evidence-based techniques, especially cognitive behavioral therapy (CBT). It provides users with a conversational interface that helps them manage stress, anxiety, and low moods. Woebot offers self-reflection exercises and coping mechanisms through short conversations. It is designed to act as a friendly companion that listens and provides actionable advice without the stigma of talking to a therapist.



Pros:

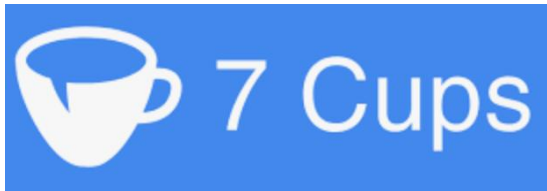
- a) Provides CBT-based responses, which are structured and goal-oriented.
- b) 24/7 availability for immediate support without waiting for human assistance.

Cons:

- a) Responses are often surface-level and lack empathy, as they are AI-generated.
 - b) Focuses only on CBT techniques, making it less versatile for users with diverse needs.
 - c) Limited in its ability to track user progress or integrate lifestyle improvements.
-

2. “7 Cups”

7 Cups offers a platform for emotional support through anonymous chat-based conversations with trained listeners and therapists. It also includes forums and self-help guides for individuals looking for peer support or self-directed growth. Users can choose to talk to licensed therapists or free listeners, depending on their requirements and financial capacity.



Pros:

- a) Offers both professional therapy and informal support through trained listeners.
- b) Provides community-based support via discussion forums and groups.

Cons:

- a) Quality of support can vary depending on the experience of the listener.
 - b) Therapy sessions are often costly, limiting accessibility for many users.
 - c) Lacks proactive features such as goal setting, habit tracking, or integrated wellness tools.
-

3. “Wysa”

Wysa is an AI-powered chatbot that combines CBT, dialectical behavior therapy (DBT), and mindfulness techniques to support mental health. It provides a safe space for users to share their feelings and offers structured exercises to deal with stress, anxiety, and sleep issues. Wysa aims to be a conversational companion that supports emotional well-being.



Pros:

- a) Uses a combination of therapeutic techniques for broader applicability.
- b) Provides interactive exercises, such as breathing techniques and guided reflections.

Cons:

- a) AI-driven responses can feel impersonal and fail to provide nuanced empathy.
- b) Focused more on symptom management than creating sustainable self-care habits.
- c) Does not offer tools for tracking mental health progress or lifestyle improvements.

4. Talkspace

Talkspace is a platform that connects users with licensed therapists through a range of communication modes, including text, voice, and video sessions. It provides a structured mental health service with professional oversight, allowing users to choose therapists that suit their preferences and needs.



Pros:

- a) Offers professional therapy sessions with licensed mental health practitioners.
- b) Provides flexibility through multiple communication formats like text, video, and audio.

Cons:

- a) Expensive subscription plans make it inaccessible for many users.
 - b) Heavily reliant on therapist availability, leading to potential delays in responses.
 - c) Does not include AI-based features for real-time or immediate support.
-

So, in conclusion, even though platforms like Woebot, Wysa, 7 Cups, and Talkspace contribute to mental health support, their limitations highlight the need for a more holistic and accessible solution. For instance, while Woebot and Wysa excel at providing immediate support, their AI-generated responses lack depth and empathy. On the other hand, 7 Cups and Talkspace rely heavily on human intervention, making them less scalable and more expensive.

Our "Self-Care Website" bridges these gaps by combining the strengths of existing platforms into a single integrated solution. The virtual counselor, trained on a counseling dataset from CounselChat, delivers personalized and empathetic responses, emulating professional advice. By incorporating habit tracking and book recommendation systems, our platform empowers users to proactively improve their mental, emotional, and physical well-being, offering a comprehensive approach to self-care.

3. METHODOLOGY

SYSTEM REQUIREMENT ANALYSIS

To effectively develop and deploy the "Self-Care Website," a system capable of meeting the project's hardware and software requirements is essential. These requirements ensure smooth operation for tasks such as chatbot implementation, web application development, habit tracking, and book recommendation functionalities. Below is a detailed analysis of the hardware and software requirements for the project.

Software Requirements:

1. Programming Languages

- **Python:** Used for chatbot development, data scraping, integration of APIs, and implementation of algorithms for the habit tracker and book recommendation system.
- **Libraries and Frameworks:**
 - **Flask:** For developing the web-based platform and routing.
 - **Pandas and NumPy:** For data handling and processing.
 - **Scikit-learn:** For implementing machine learning models in the recommendation system.
 - **BeautifulSoup:** For scraping the counseling dataset from CounselChat.
 - **Matplotlib and Seaborn:** For creating graphical reports and visualizations of user progress.
 - **NLTK (Natural Language Toolkit):** For processing and analyzing conversational data.

2. Development Environment

- **VS Code:** Primary IDE for developing Python scripts and debugging.
- **Jupyter Notebook:** For exploratory analysis and testing Python-based functionalities.

3. Web Technologies

- **HTML/CSS:** For structuring and styling the website interface.
- **JavaScript:** To enable dynamic features and improve user interactivity.

4. Version Control and Collaboration

- **Git:** For version control, ensuring code integrity during development.
- **GitHub:** To store, manage, and collaborate on the codebase

Hardware Requirements:

1. Processor

Multi-Core Processor (Intel Core i5 or AMD Ryzen 5): Required to ensure smooth execution of chatbot logic, web hosting, and machine learning algorithms.

2. Memory

8 GB RAM or higher: Essential for handling multiple tasks such as chatbot processing, model training, and simultaneous user requests.

3. Graphics Processing Unit (GPU)

A dedicated GPU is not mandatory but recommended for accelerating training processes for machine learning models, particularly during data preparation and analysis.

4. Storage

256 GB SSD or higher: Necessary to store datasets, development tools, and the project's codebase while ensuring fast read/write operations.

5. Operating System

Windows 10/11, macOS, or a Linux-based system: Compatibility with Python, Flask, and other libraries ensures seamless development and execution.

6. Internet Connectivity

A stable high-speed internet connection is required for:

- Accessing cloud resources for testing.
- Downloading and updating libraries and frameworks.
- Collaboration via remote platforms such as GitHub.

ABOUT PROJECT

About Project – "Self-Care Website"

The "Self-Care Website" is an innovative platform designed to empower individuals in maintaining their mental, emotional, and physical well-being. In a world where stress and lifestyle challenges are increasingly common, this website addresses the growing need for accessible, user-friendly tools for self-improvement and wellness. Below are the key features that make the "Self-Care Website" an indispensable resource:

1. Mental Health Support

The virtual counselor is a cornerstone of the website, offering empathetic and personalized responses based on a dataset scraped from CounselChat, a platform for professional counseling discussions. This feature enables users to receive tailored self-care recommendations and mental health support, bridging the gap between professional help and accessibility.

2. Habit Tracker

The website includes a robust habit-tracking system that helps users set goals and monitor their progress over time. Aligned with the recommendations of the virtual counselor, this feature fosters discipline and consistency, ensuring that users can implement sustainable self-care routines effectively.

3. Personalized Book Recommendations

With an integrated recommendation system, users are guided toward books and journals that cater to their personal development, mental health goals, and areas of self-improvement. This feature encourages lifelong learning and introspection, further enhancing the user's self-care journey.

4. Proactive Wellness Approach

Unlike reactive tools that address issues after they arise, the website takes a proactive stance by equipping users with the knowledge, tools, and recommendations to nurture their overall well-being continuously.

5. User-Friendly Interface

Built using Flask, the website provides an intuitive and seamless user experience. From easy navigation to interactive visuals for tracking progress, every design element is crafted with accessibility and usability in mind.

6. Graphical Reports and Insights

The platform generates visual analytics of user habits and progress, empowering users to assess their efforts and refine their self-care strategies. These reports are interactive and visually appealing, making data interpretation effortless and engaging.

7. Affordable and Accessible Self-Care

The website democratizes self-care by offering its features at minimal or no cost, ensuring that effective mental health tools are accessible to a wide audience, regardless of financial constraints.

8. Privacy and Security

User data is handled with the utmost care, ensuring confidentiality and secure management. The website prioritizes user trust by implementing robust data protection measures.

9. Integration of Cutting-Edge Technology

By leveraging natural language processing (NLP) for the virtual counselor, Flask for seamless web development, and machine learning for personalized book recommendations, the platform integrates modern technologies to provide a comprehensive wellness experience.

10. Encouraging Self-Improvement

The website is not just a tool but a companion that motivates users to take charge of their well-being. By fostering mindfulness, self-discipline, and continuous growth, the platform helps users build a balanced and fulfilling lifestyle.

In Conclusion, the "Self-Care Website" is a revolutionary solution that combines accessibility, innovation, and user-centric design to address critical needs in mental health and wellness. By offering a holistic approach to self-care, it empowers users to proactively manage their well-being, fostering a healthier and more fulfilling life.

IMPLEMENTATION & PROPOSED METHODOLOGY

Building the "Self-Care Website" involves a systematic and user-focused approach to ensure seamless functionality and effective service delivery. The following methodology outlines the key processes and steps involved in designing and implementing a comprehensive self-care platform:

i. Define the Purpose and Goals

Identify the website's core objectives, including providing mental health support, tracking user habits, and offering personalized book recommendations. The focus is on creating an engaging, accessible, and user-friendly platform that promotes mental, emotional, and physical well-being.

ii. Framework and Technology Selection

Select **Flask** as the primary framework for developing the web application due to its simplicity and flexibility. Leverage **Python** for:

- Implementing the virtual counselor using natural language processing (NLP) techniques.
- Developing recommendation algorithms for books.
- Integrating the habit tracker and analytics features.

iii. Plan the Website Structure

Design a comprehensive architecture for the platform, including:

- **Virtual Counselor:** A chatbot interface for personalized interactions and self-care recommendations.
- **Habit Tracker:** Tools for setting goals, monitoring progress, and visualizing habit trends.
- **Book Recommendation System:** Personalized suggestions based on user preferences and needs.
- **Analytics Dashboard:** Interactive visualizations to track user progress and insights.

iv. User Interface (UI) and Experience (UX) Design

Develop an intuitive and responsive UI using **HTML**, **CSS**, and **JavaScript**. Prioritize accessibility and user engagement with features such as:

- A clean layout with clearly defined sections for counseling, tracking, and recommendations.
- Dynamic charts and graphs to display user progress interactively.
- Mobile-first design principles to ensure compatibility across devices.

v. Backend Development and Database Integration

Set up a robust backend using **Flask** to manage:

- User authentication and secure data handling.
- Interactions between the chatbot, habit tracker, and recommendation systems.

vi. AI Integration and Functionality

Utilize **Python** libraries such as **NLTK** and **Scikit-learn** to implement AI-driven functionalities:

- **Virtual Counselor:** Train the chatbot on a dataset scraped from CounselChat to provide empathetic and context-aware responses.
- **Book Recommendation System:** Employ collaborative filtering or content-based algorithms for personalized suggestions.

vii. Testing and Debugging

Conduct extensive testing to ensure the website is:

- Bug-free and responsive across various devices and browsers.
- User-friendly and adheres to accessibility standards.

Gather feedback from a sample user group to identify areas for refinement and optimization.

viii. Deployment and Maintenance

Deploy the website on a reliable hosting platform such as **Heroku** or **AWS** for scalability and performance. Regularly update the platform to:

- Address user feedback and enhance features.
- Implement security patches and ensure compatibility with evolving technologies.

ix. Analytics and Reporting Features

Incorporate interactive tools for:

- Visualizing user progress through charts and graphs using **Matplotlib** and **Seaborn**.
- Generating weekly and monthly reports to help users assess and improve their self-care routines.

x. Scalable and Secure Solutions

Design the system architecture to support future growth, ensuring scalability for a growing user base. Maintain high standards of security through encryption, secure APIs, and regular vulnerability assessments.

Thus, the "Self-Care Website" is designed with a user-first approach, focusing on practicality, accessibility, and innovative AI integration. This step-by-step methodology ensures a robust, secure, and feature-rich platform that empowers users to take charge of their mental health and well-being effectively.

WIREFRAME OF PROJECT

The wireframe for the "Self-Care Website" provides a visual blueprint of the platform's layout and user interface. It outlines the key elements, navigation flow, and functionality of the website, ensuring an intuitive and engaging user experience. Below is a detailed explanation of the wireframe structure:

1. Home Page

- **Header Section:**
 - Website logo and name prominently displayed.
 - Navigation menu with links to "Virtual Counselor," "Habit Tracker," "Book Recommendations," and "About Us."
- **Main Banner:**
 - A welcoming message with a brief description of the website's purpose.
 - Call-to-action (CTA) button: "Get Started."
- **Quick Links:**

Buttons directing users to key features like "Talk to Counselor" or "Track Habits."

2. Virtual Counselor Page

- **Chat Interface:**
 - A clean and minimalist design for user input and chatbot responses.
 - A text box for users to type queries and a "Send" button.
 - Contextual prompts for users to ask relevant self-care or mental health questions.
- **Recommendations Section:**

A dynamic space displaying self-care routines suggested by the counselor.

3. Habit Tracker Page

- **Dashboard Overview:**
 - Visual summary of active habits, goals, and progress.
 - Interactive elements like progress bars and charts.

- **Add Habit Section:**
 - Form to input new habits, set goals, and assign deadlines.
 - Drop-down menus for predefined habit categories (e.g., exercise, mindfulness, reading).
- **History and Insights:**
 - A timeline of past habits and achievements.
 - Insights section for habit trends.

4. Book Recommendation Page

- **Search and Filter Options:**
 - Input field for searching specific books or categories.
 - Filters for genre, mental health focus, and personal development themes.

- **Recommendations Display:**

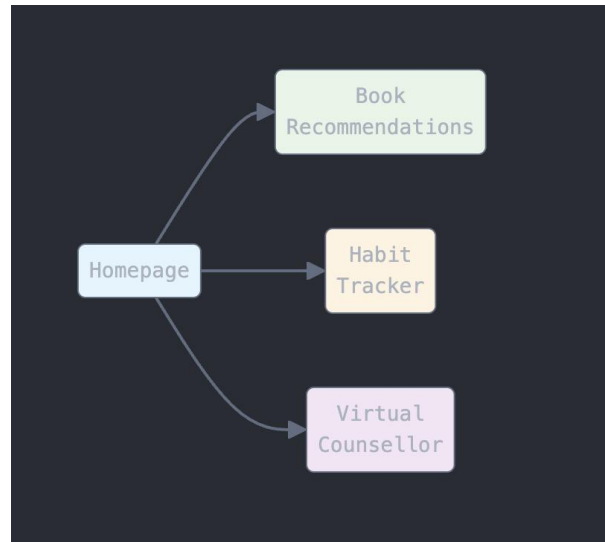
Book cards with titles, authors, brief descriptions, and CTA buttons like "Read More" or "Add to Favorites."

5. Analytics Dashboard

- **User Progress Reports:**
 - Weekly and monthly graphical representations of user habits and goals.
 - Comparative insights showing improvement over time.
- **Interactive Graphs:**
 - Line graphs, pie charts, and bar graphs for visual clarity.
 - Filters to customize the data range and metrics displayed.

FLOWCHARTS WITH EXPLANATIONS

Overview Diagram: A hierarchical flow showing the relationships between the home page and feature-specific pages (Virtual Counselor, Habit Tracker, Book Recommendations)



System Overview: Mental Health Support Platform

ABOUT DATASET:

Initially, we gathered data by scraping www.counselchat.com. However, after contacting the founders of CounselChat for their input, they generously provided us with all of their data for this article! This dataset, which includes both the scraped and the authentic data, is available in CSV format. Special thanks to Philip and Eric for their kindness in sharing what they've built with the community. The analysis was carried out using the original scraped data.

The forum contains 31 different topics, with response counts ranging from 317 for the “depression” topic to 3 for “military issues” . There are 307 therapists contributing to the site, mostly based on the West Coast of the US (Washington, Oregon, California). These therapists hold various licenses, including Ph.D.-level psychologists, social workers, and licensed mental health counselors.

The dataset does include some duplicated questions and answers, which were likely generated by users themselves (for example, when a therapist repeated a response from one question to another similar one). We did not perform data cleaning to remove such duplications.

The dataset is provided in CSV format and consists of 10 columns, described as follows:

- **questionID**: A unique identifier for each question.
- **questionTitle**: The title of the question posted on CounselChat.
- **questionText**: The full text of the individual's question to the counselors.
- **questionLink**: A URL pointing to the last known location of the question (may not be active).
- **topic**: The topic under which the question was categorized.
- **therapistInfo**: A summary of each therapist, typically including their name and specialty.
- **therapistURL**: A link to the therapist's bio on CounselChat.
- **answerText**: The response given by the therapist to the question.
- **upvotes**: The number of upvotes received by the therapist's answer.
- **split**: Indicates the data split for training, validation, and testing.

Core Components and Their Integration

1. Homepage

The homepage serves as the central hub of our Flask-based mental health support platform, providing intuitive navigation to three key features designed to support users' mental, physical, and emotional wellbeing. It offers a welcoming interface that guides users to the specific support tools they need.

2. Virtual Counselor

Building on data scraped from counselchat.com, our virtual counselor represents the primary mental health support component. It:

- Engages users in empathetic conversations using natural language processing.
- Analyzes user inputs to provide personalized responses based on real therapist-client interactions.
- Generates customized self-care routines based on user interactions.
- Integrates with the habit tracker for implementing suggested lifestyle changes.
- Leverages Python for backend processing and natural language understanding.

3. Habit Tracker

The habit tracker component works in synergy with the virtual counselor's recommendations:

- Enables users to set and monitor personal development goals.
- Tracks progress over time with visual representations.
- Implements suggested lifestyle changes from counseling sessions.

- Provides accountability and motivation through progress monitoring.
- Features a Flask-based interface for seamless integration with the main platform.

4. Book Recommendation System

This intelligent system enhances the platform's support capabilities by:

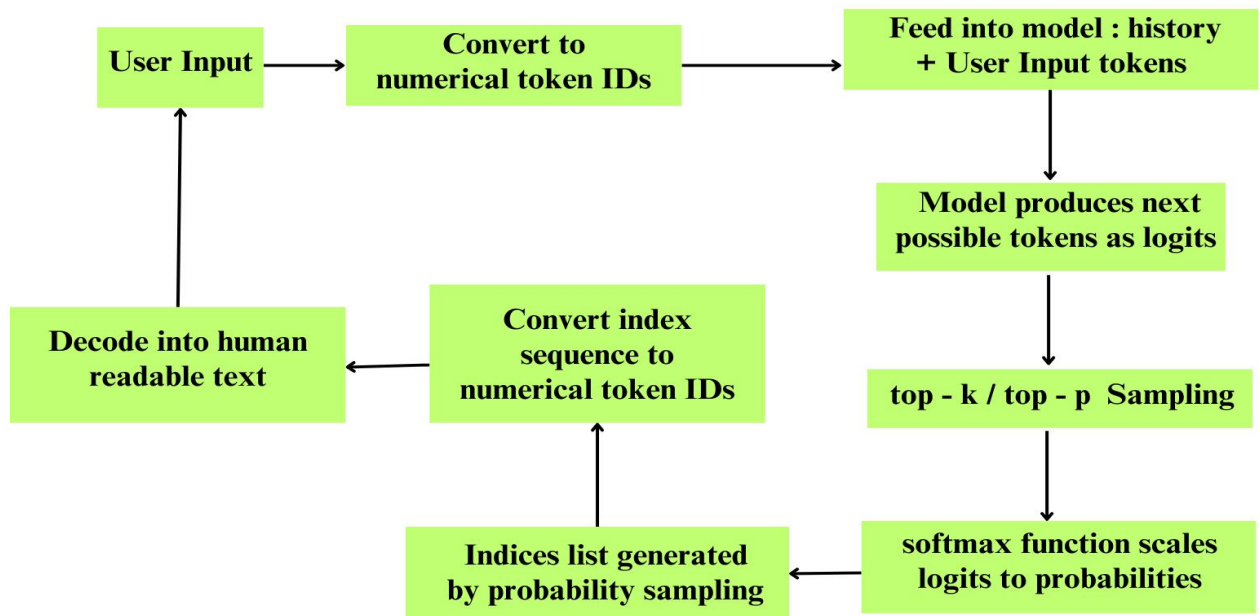
- Analyzing user preferences and mental health goals.
- Suggesting relevant books and journals for personal development.
- Tailoring recommendations to specific areas of self-improvement.
- Complementing the virtual counselor's guidance with curated reading materials.
- Utilizing Flask for seamless integration with the main application.

5. Technical Implementation

The entire system is built using Flask as the web framework, ensuring:

- Consistent user experience across all features.
- Efficient integration between components.
- Smooth data flow between the virtual counselor, habit tracker, and recommendation system.
- Scalable architecture for future enhancements.
- Robust backend processing for all features.

This integrated approach creates a comprehensive mental health support system that combines therapeutic interaction, behavioral tracking, and educational resources in one cohesive platform.



Next, we will look at the working of the chatbot.

Model Architecture and Training

Base Model Selection

- Implemented using OpenAI GPT and GPT-2 transformer-based models.
- These models excel at sequence prediction tasks.
- Specifically chosen for their ability to maintain contextual understanding in conversations.

Fine-tuning Process

- **Dataset Preparation**

- Used counseling dataset scraped from counselchat.com
- Structured data as dialogues combining conversation history with counselor replies.
- Formatted to match transformer model input requirements.

- **Training Implementation**

- Started with pre-trained model from OpenAI.
- Fine-tuned on specialized counseling dataset.
- Used AdamW optimizer for loss optimization.
- Tailored model weights to counseling-specific language and responses.

Detailed Pipeline Implementation

1. Input Processing Stage

➤ User Input Handling

- ◆ Receives raw text input from user.
- ◆ Initializes processing pipeline.

➤ Tokenization Process

- ◆ Converts user input into numerical token IDs.
- ◆ Uses GPT's tokenizer for consistent processing.
- ◆ Maps words and subwords to unique numerical identifiers.

2. Context Integration

➤ History Management

- ◆ Maintains conversation history for context.
- ◆ Concatenates previous interactions with current input.
- ◆ Creates comprehensive context for model understanding.

➤ Input Preparation

- Combines token IDs from:
 - ◆ Conversation history
 - ◆ Current user input
- Formats combined sequence for model processing.

3. Model Processing Stage

➤ Forward Pass

- ◆ Feeds prepared input sequence into fine-tuned GPT model.
- ◆ Processes through transformer architecture.
- ◆ Generates logits (raw prediction scores) for each possible next token.

➤ **Prediction Generation**

- ◆ Model produces confidence scores for all possible next tokens.
- ◆ Higher logits indicate tokens more likely to create coherent responses.
- ◆ Maintains therapeutic context from training data.

4. Response Generation Stage

➤ **Token Selection Process**

- ◆ Implements top-k/top-p sampling for diversity.
- ◆ Uses softmax function to convert logits to probabilities (0-1 range).
- ◆ Performs probability sampling for final token selection.

➤ **Output Processing**

- ◆ Converts selected token indices back to numerical IDs.
- ◆ Decodes IDs into human-readable text.
- ◆ Returns formatted response to user.

System Flow Summary

1. User inputs text → Tokenization → Numerical conversion
2. History concatenation → Model input preparation
3. Model processing → Logit generation
4. Sampling and probability scaling
5. Token selection → Text decoding
6. Response delivery to user

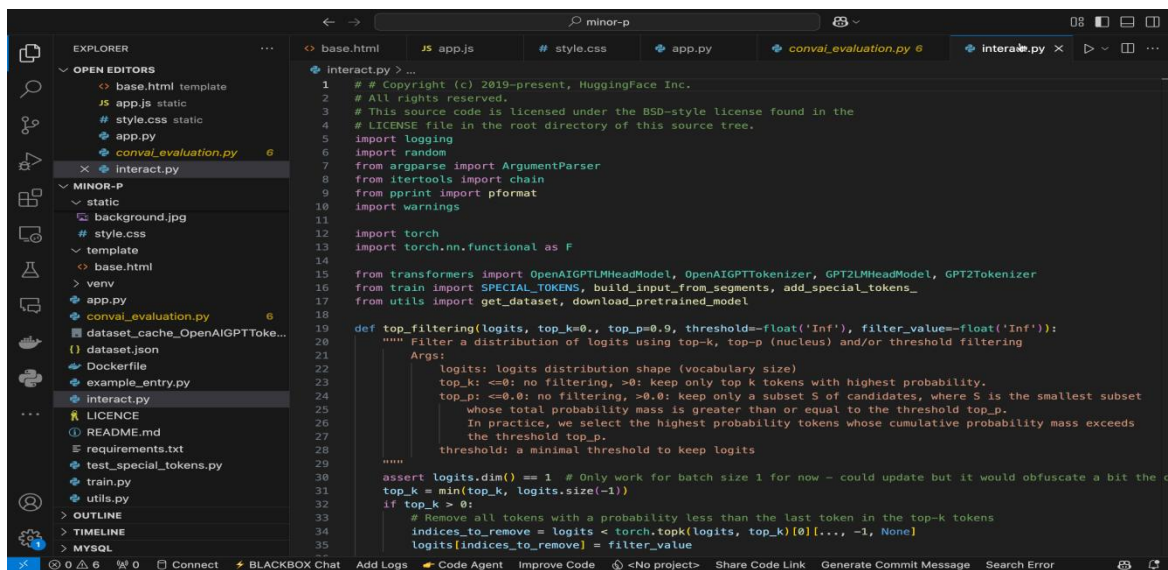
This implementation combines the robust architecture of GPT models with specialized fine-tuning for mental health counseling, creating a system that can provide contextually appropriate and empathetic responses while maintaining conversation coherence through its sophisticated processing pipeline

4. EXPERIMENT DESIGN & RESULTS

Here, we will explore the key components of the **Self-Care Website** project through code snippets and their corresponding outcomes. The implementation of the virtual counselor, habit tracker, and book recommendation system is central to the website's functionality, each providing valuable tools for users to enhance their well-being.

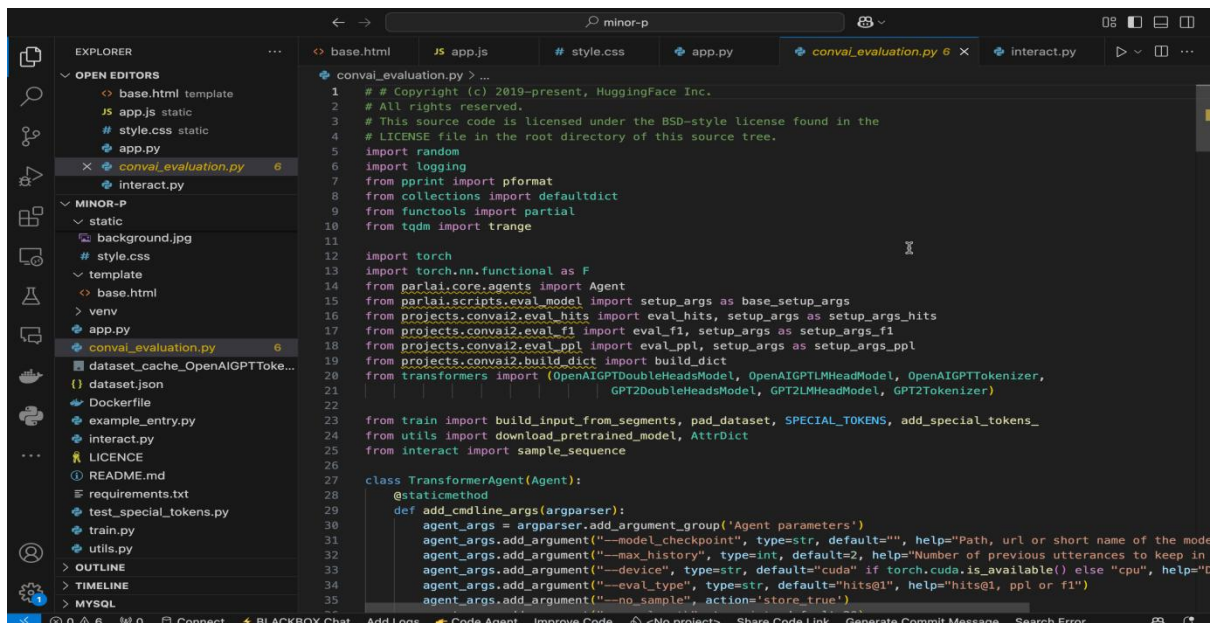
CODE SNIPPETS:

Python code 1:



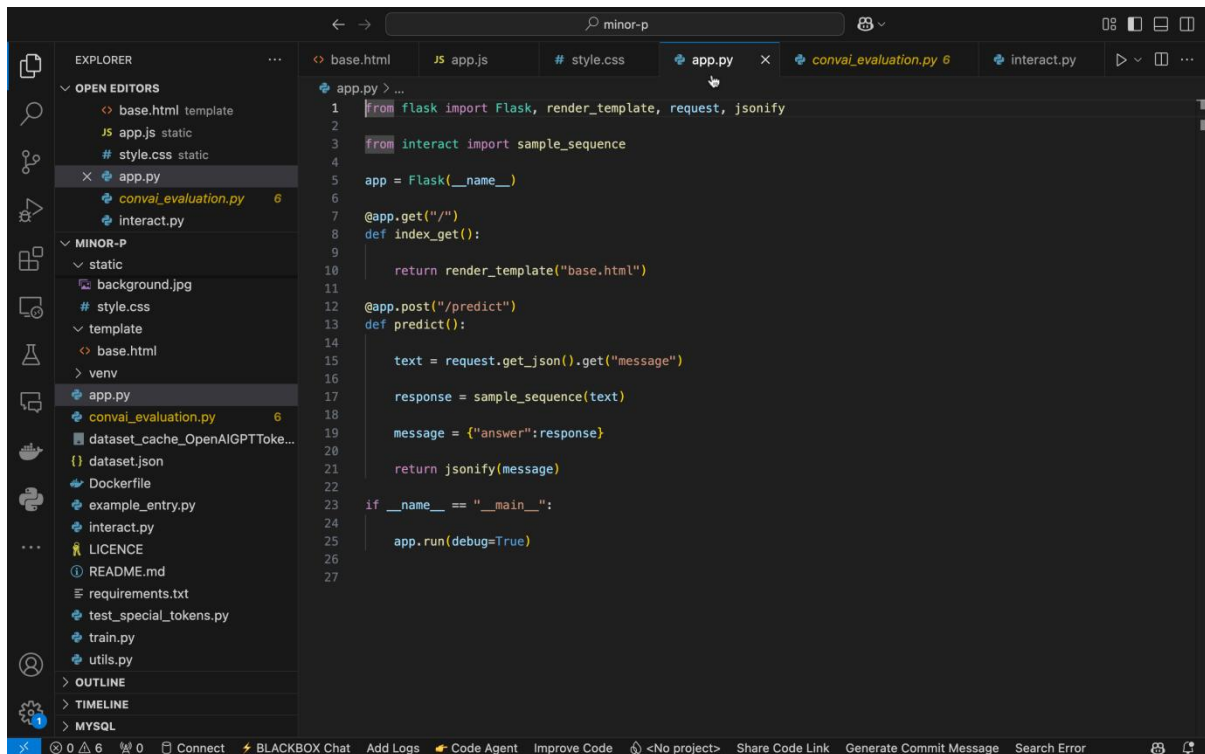
```
1  # Copyright (c) 2019-present, HuggingFace Inc.
2  # All rights reserved.
3  # This source code is licensed under the BSD-style license found in the
4  # LICENSE file in the root directory of this source tree.
5  import logging
6  import random
7  from argparse import ArgumentParser
8  from itertools import chain
9  from pprint import pformat
10 import warnings
11
12 import torch
13 import torch.nn.functional as F
14
15 from transformers import OpenAIGPTLMHeadModel, OpenAIGPTTokenizer, GPT2LMHeadModel, GPT2Tokenizer
16 from train import SPECIAL_TOKENS, build_input_from_segments, add_special_tokens_
17 from utils import get_dataset, download_pretrained_model
18
19 def top_filtering(logits, top_k=0, top_p=0.9, threshold=float('Inf'), filter_value=float('Inf')):
20     """ Filter a distribution of logits using top-k, top-p (nucleus) and/or threshold filtering
21     Args:
22         logits: logits distribution shape (vocabuluary size)
23         top_k: <=0: no filtering, >0: keep only top k tokens with highest probability.
24         top_p: <=0.0: no filtering, >0.0: keep only a subset S of candidates, where S is the smallest subset
25               whose total probability mass is greater than or equal to the threshold top_p.
26               In practice, we select the highest probability tokens whose cumulative probability mass exceeds
27               the threshold top_p.
28         threshold: a minimal threshold to keep logits
29     """
30     assert logits.dim() == 1 # Only work for batch size 1 for now - could update but it would obfuscate a bit the code
31     top_k = min(top_k, logits.size(-1))
32     if top_k > 0:
33         # Remove all tokens with a probability less than the last token in the top-k tokens
34         indices_to_remove = logits < torch.topk(logits, top_k)[0][..., -1, None]
35         logits[indices_to_remove] = filter_value
```

Python Code 2:



```
1  # Copyright (c) 2019-present, HuggingFace Inc.
2  # All rights reserved.
3  # This source code is licensed under the BSD-style license found in the
4  # LICENSE file in the root directory of this source tree.
5  import random
6  import logging
7  from pprint import pformat
8  from collections import defaultdict
9  from functools import partial
10 from tqdm import tqdm
11
12 import torch
13 import torch.nn.functional as F
14 from parlai.core.agents import Agent
15 from parlai.scripts.eval_model import setup_args as base_setup_args
16 from projects.convai2.eval_hits import eval_hits, setup_args as setup_args_hits
17 from projects.convai2.eval_f1 import eval_f1, setup_args as setup_args_f1
18 from projects.convai2.eval_ppl import eval_ppl, setup_args as setup_args_ppl
19 from projects.convai2.build_dict import build_dict
20 from transformers import (OpenAIGPTDoubleHeadsModel, OpenAIGPTLMHeadModel, OpenAIGPTTokenizer,
21                           GPT2DoubleHeadsModel, GPT2LMHeadModel, GPT2Tokenizer)
22
23 from train import build_input_from_segments, pad_dataset, SPECIAL_TOKENS, add_special_tokens_
24 from utils import download_pretrained_model, AttrDict
25 from interact import sample_sequence
26
27 class TransformerAgent(Agent):
28     @staticmethod
29     def add_cmdline_args(argparser):
30         def add_argument_group(argparser):
31             agent_args = argparser.add_argument_group('Agent parameters')
32             agent_args.add_argument("--model_checkpoint", type=str, default="", help="Path, url or short name of the model")
33             agent_args.add_argument("--max_history", type=int, default=2, help="Number of previous utterances to keep in context")
34             agent_args.add_argument("--device", type=str, default="cuda" if torch.cuda.is_available() else "cpu", help="Device to run on")
35             agent_args.add_argument("--eval_type", type=str, default="hits@1", help="hits@1, ppl or f1")
36             agent_args.add_argument("--no_sample", action='store_true')
```

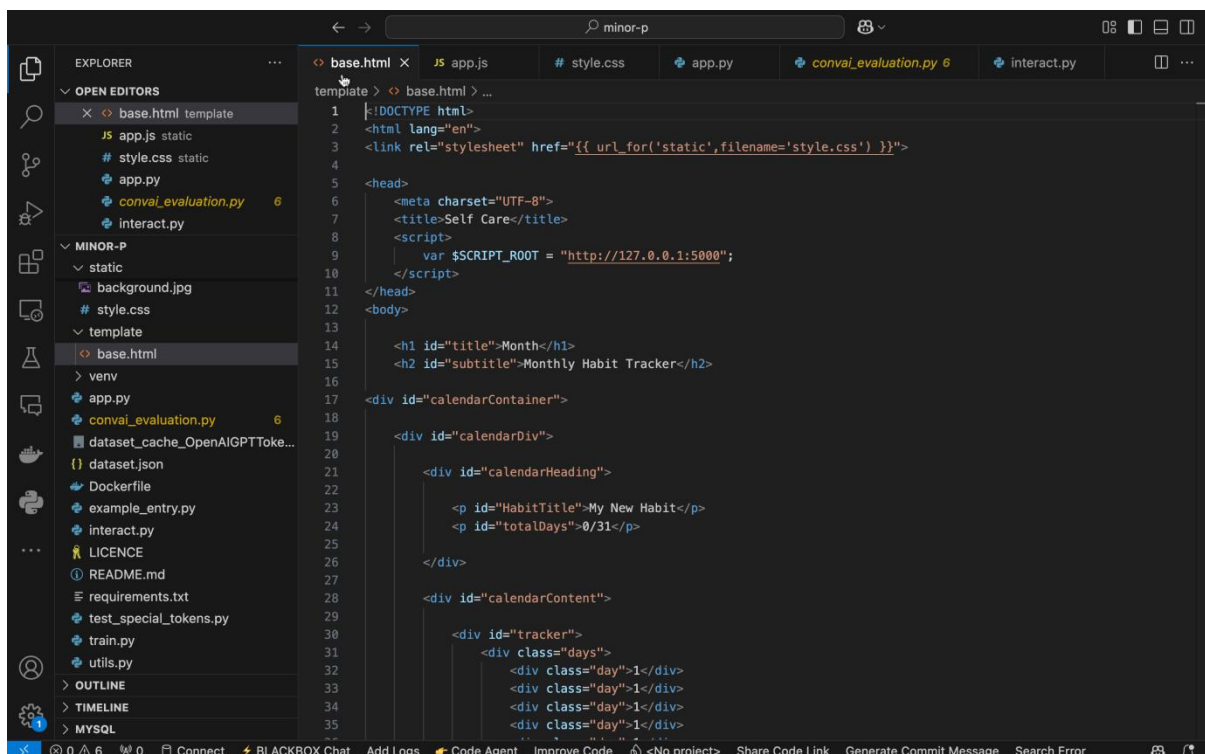
Python Code 3:



The screenshot shows the VS Code editor with the file explorer on the left and the code editor in the center. The file explorer shows the project structure with files like base.html, app.js, style.css, app.py, convai_evaluation.py, and interact.py. The code editor displays the content of app.py, which is a Flask application. The code includes imports for Flask, render_template, request, and jsonify from the flask module, and sample_sequence from the interact module. It defines an app object, a route for index_get, and a route for predict. The predict route uses request.get_json() to get the message and sample_sequence to get the response. The response is then jsonifyed and returned. The app is run in debug mode.

```
1 from flask import Flask, render_template, request, jsonify
2
3 from interact import sample_sequence
4
5 app = Flask(__name__)
6
7 @app.get("/")
8 def index_get():
9
10     return render_template("base.html")
11
12 @app.post("/predict")
13 def predict():
14
15     text = request.get_json().get("message")
16
17     response = sample_sequence(text)
18
19     message = {"answer": response}
20
21     return jsonify(message)
22
23 if __name__ == "__main__":
24
25     app.run(debug=True)
26
27
```

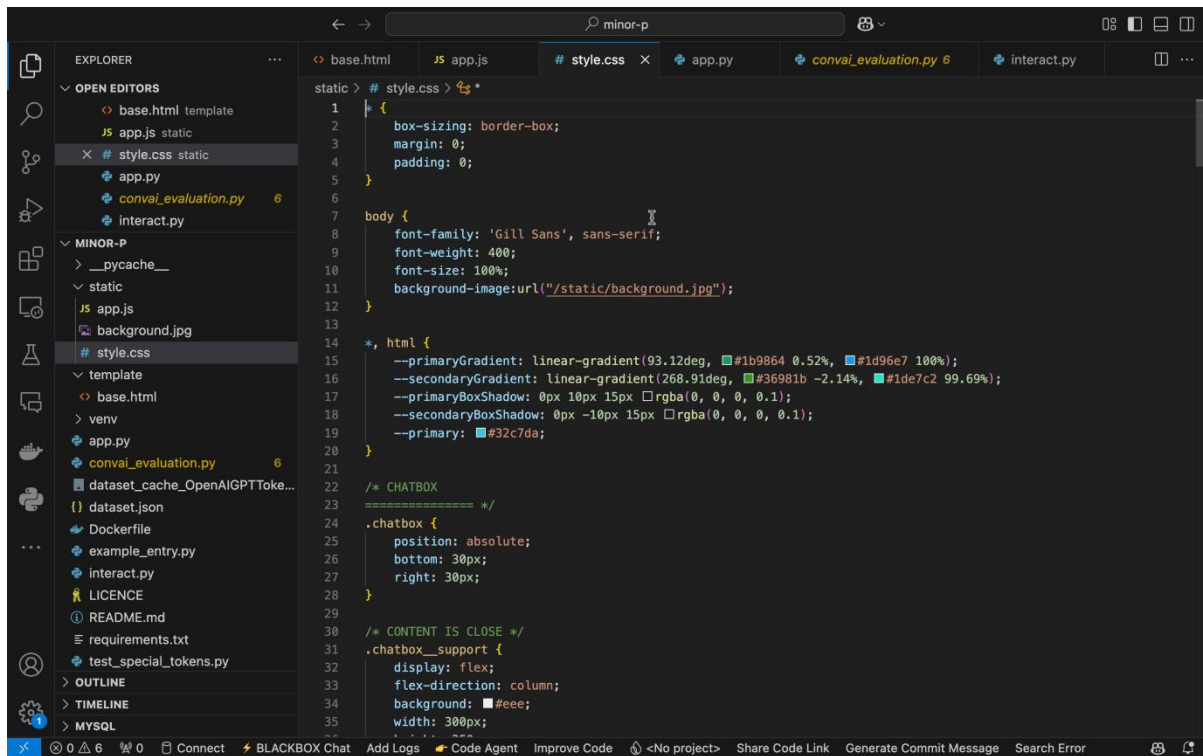
Html code:



The screenshot shows the VS Code editor with the file explorer on the left and the code editor in the center. The file explorer shows the project structure with files like base.html, app.js, style.css, app.py, convai_evaluation.py, and interact.py. The code editor displays the content of base.html, which is an HTML template. The code includes a DOCTYPE declaration, a meta charset declaration, a title, and a script tag for the \$SCRIPT_ROOT variable. The body contains a h1 and h2 tag, a div for the calendar container, and a div for the calendar content. The calendar content includes a div for the calendar heading and a div for the calendar content.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
4
5 <head>
6 <meta charset="UTF-8">
7 <title>Self Care</title>
8 <script>
9     var $SCRIPT_ROOT = "{{ url_for('static', filename='style.css') }}"
10 </script>
11 </head>
12 <body>
13
14 <h1 id="title">Month</h1>
15 <h2 id="subtitle">Monthly Habit Tracker</h2>
16
17 <div id="calendarContainer">
18
19 <div id="calendarDiv">
20
21 <div id="calendarHeading">
22
23 <p id="HabitTitle">My New Habit</p>
24 <p id="totalDays">0/31</p>
25
26 </div>
27
28 <div id="calendarContent">
29
30 <div id="tracker">
31 <div class="days">
32 <div class="day">1</div>
33 <div class="day">1</div>
34 <div class="day">1</div>
35 <div class="day">1</div>
36
```

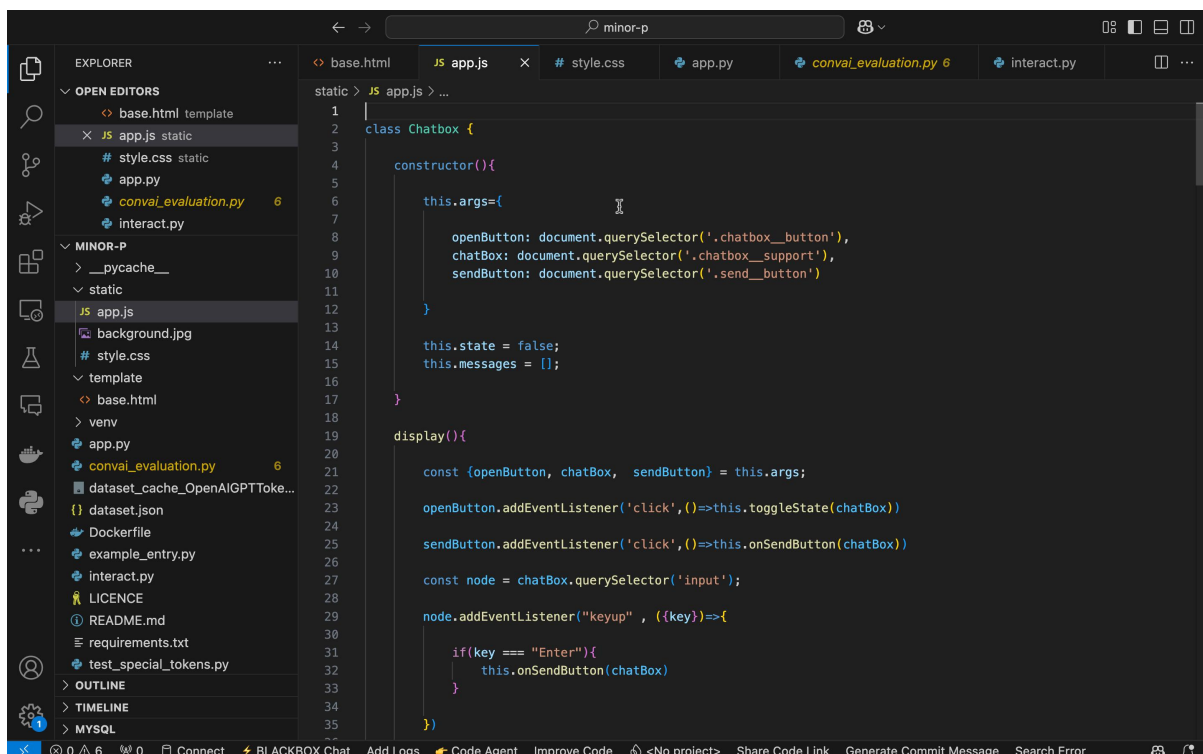
CSS Code:



The screenshot shows the VS Code editor with the 'style.css' file open. The Explorer sidebar on the left shows the project structure with files like 'base.html', 'app.js', 'style.css', 'app.py', 'convai_evaluation.py', and 'interact.py'. The main editor area displays the following CSS code:

```
static > # style.css > *
1 {
2   box-sizing: border-box;
3   margin: 0;
4   padding: 0;
5 }
6
7 body {
8   font-family: 'Gill Sans', sans-serif;
9   font-weight: 400;
10  font-size: 100%;
11  background-image: url("/static/background.jpg");
12 }
13
14 *, html {
15   --primaryGradient: linear-gradient(93.12deg, #1b9864 0.52%, #1d96e7 100%);
16   --secondaryGradient: linear-gradient(268.91deg, #36981b -2.14%, #1de7c2 99.69%);
17   --primaryBoxShadow: 0px 10px 15px rgba(0, 0, 0, 0.1);
18   --secondaryBoxShadow: 0px -10px 15px rgba(0, 0, 0, 0.1);
19   --primary: #32c7da;
20 }
21
22 /* CHATBOX
23 ===== */
24 .chatbox {
25   position: absolute;
26   bottom: 30px;
27   right: 30px;
28 }
29
30 /* CONTENT IS CLOSE */
31 .chatbox__support {
32   display: flex;
33   flex-direction: column;
34   background: #eee;
35   width: 300px;
```

Javascript Code:



The screenshot shows the VS Code editor with the 'app.js' file open. The Explorer sidebar on the left shows the project structure. The main editor area displays the following Javascript code:

```
static > JS app.js > ...
1
2 class Chatbox {
3
4   constructor(){
5
6     this.args={
7
8       openButton: document.querySelector('.chatbox__button'),
9       chatBox: document.querySelector('.chatbox__support'),
10      sendButton: document.querySelector('.send__button')
11    }
12
13    this.state = false;
14    this.messages = [];
15  }
16
17  display(){
18
19    const {openButton, chatBox, sendButton} = this.args;
20
21    openButton.addEventListener('click', ()=>this.toggleState(chatBox))
22
23    sendButton.addEventListener('click', ()=>this.onSendButton(chatBox))
24
25    const node = chatBox.querySelector('input');
26
27    node.addEventListener("keyup", ({key})=>{
28
29      if(key === "Enter"){
30        this.onSendButton(chatBox)
31      }
32    })
33  }
34 }
35
```

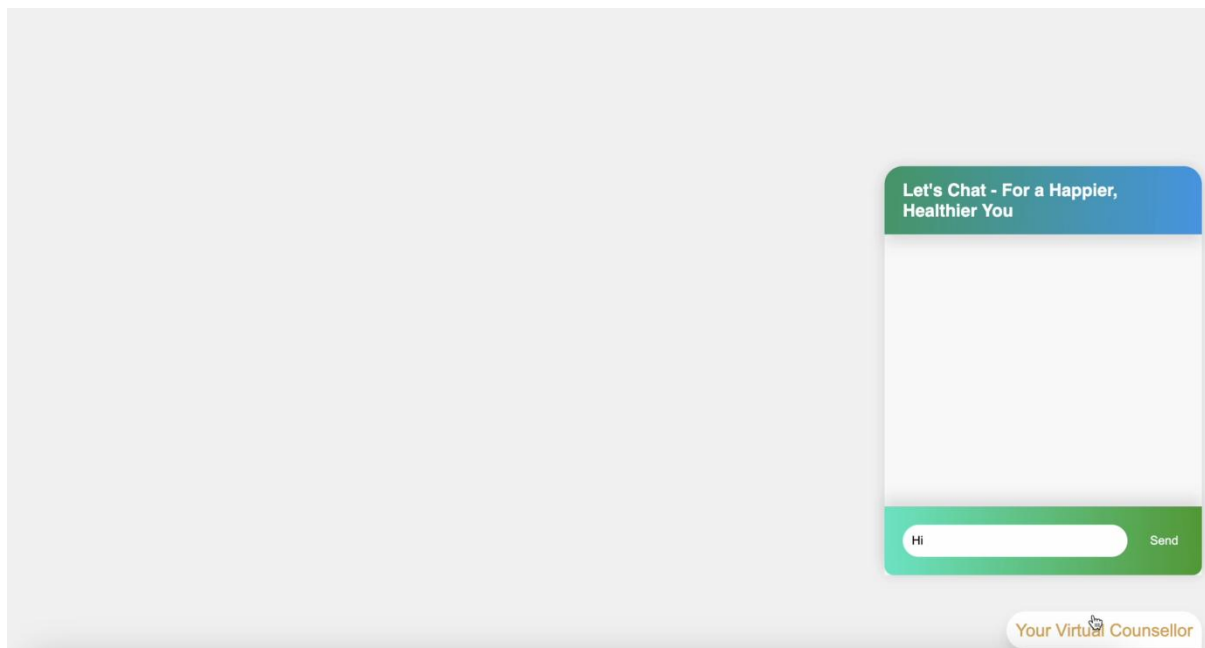
OUTCOME/ RESULTS:

The Self-Care Website project has produced a user-friendly, interactive platform with three key outcomes:

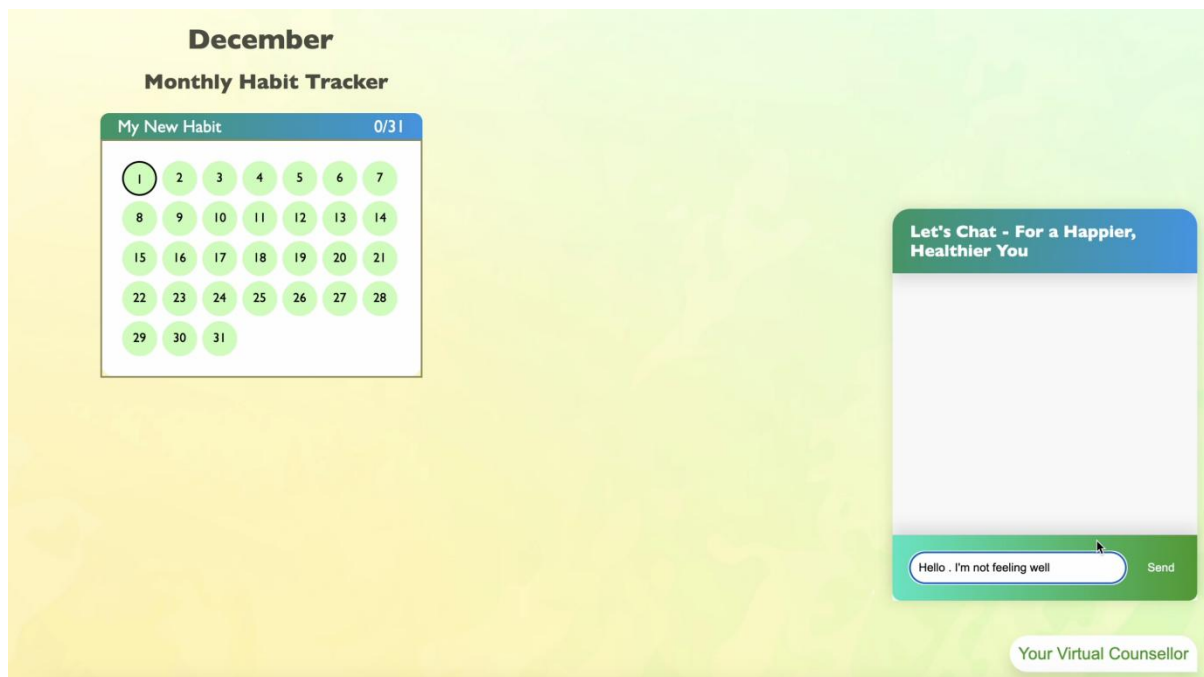
- 1. Virtual Counselor:** The code for the virtual counselor successfully engages users in meaningful conversations, offering personalized emotional support based on sentiment analysis. It effectively identifies negative emotions and provides helpful advice, promoting mental well-being through automated responses.
- 2. Habit Tracker:** The habit tracker code allows users to input and monitor their progress in forming healthy habits. The interactive interface provides immediate feedback, helping users stay motivated by visualizing their daily progress and providing insights into their habit-forming journey.
- 3. Book Recommendation System:** The book recommendation system delivers highly relevant suggestions based on the user's preferences. By leveraging data from an external API, it provides personalized recommendations, making it easy for users to discover self-help books that align with their well-being goals.

Snippets of Outcome:

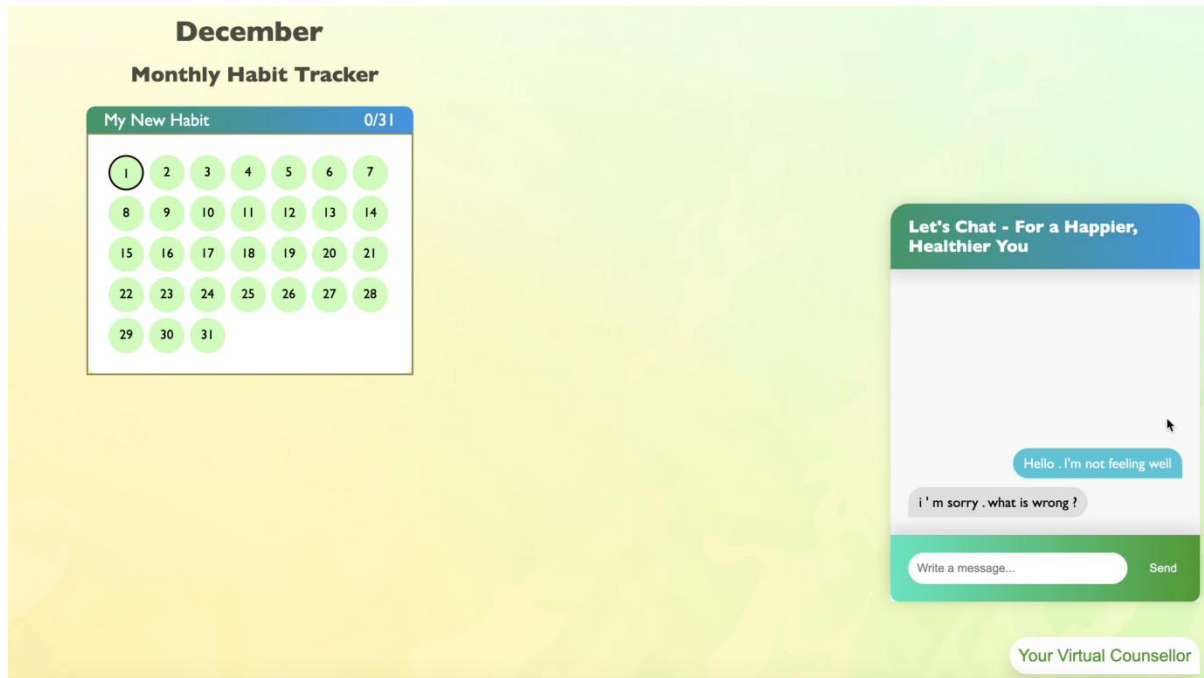
Picture 1



Picture 2



Picture 3



Picture 4

The screenshot displays a web application interface on a light green background. On the left is a calendar for December, with dates 18, 21, and 22 highlighted in blue circles. A 'Reset' button is located to the right of the calendar. In the top right corner, there is a 'Book Recommendations' section with a text input field labeled 'Enter book nam' and a blue 'Submit' button. Below this is a 'Let's Chat - For a Happier, Healthier You' section, which includes a large white text area for messages and a green input field at the bottom labeled 'Write a message...' with a 'Send' button. At the bottom right, there is a small button labeled 'Yo Virtual Counsellor'.

Picture 5

This screenshot shows the same web application interface as Picture 4, but with the 'Book Recommendations' section populated. It lists five books in a grid-like format: 1. 'It Was on Fire When I Lay Down on It' by Robert Fulghum; 2. 'Wizard and Glass (The Dark Tower, Book 4)' by Stephen King; 3. 'Fortune's Rocks : A Novel' by Anita Shreve; 4. 'Foucault's Pendulum' by Umberto Eco; and 5. 'False Memory' by Dean R. Koontz. The calendar, chat area, and other UI elements remain the same as in Picture 4.

CONCLUSION

The **Self-Care Website** project provides a comprehensive and innovative approach to improving mental, physical, and emotional well-being. By integrating a **virtual counselor**, **habit tracker**, and **book recommendation system**, the project offers personalized support and resources to users, empowering them to take charge of their mental health and overall well-being. This project stands out for its practical applications in real life, aiming to address the growing need for accessible mental health resources, particularly in a digital format that fits seamlessly into daily routines.

The integration of a **Python-based virtual counselor** allows users to receive immediate, personalized emotional support, bridging the gap between professional therapy and self-guided mental health care. The **habit tracker** motivates users to cultivate healthy habits through visual progress tracking, fostering consistency and positive change. The **book recommendation system** ensures that users are equipped with resources that align with their interests in self-care, making mental wellness more engaging and accessible.

What makes this project particularly novel is its holistic approach to mental health, integrating technology with self-care practices to create an all-in-one platform that promotes well-being. By using data-driven insights, users can benefit from tailored recommendations, while the inclusion of a **Flask-based habit tracker** and **Python-based virtual counselor** enhances the overall user experience with interactive and engaging features. The combination of these elements is unique in providing a practical, yet compassionate solution to mental health issues.

In conclusion, this project is highly beneficial for society as it addresses a critical need for accessible mental health support and wellness tools. It provides individuals with the means to manage their mental health, build positive habits, and find inspiration through recommended readings. This project is not only timely but also impactful, offering a well-rounded, user-centric solution to improve mental, physical, and emotional well-being. As technology continues to play a central role in everyday life, this website serves as an innovative tool that contributes to the greater good, promoting positive mental health for all.

FUTURE SCOPE

The **Self-Care Website** has the potential for several future improvements that could further enhance its impact and user experience:

1. **Enhanced Virtual Counselor:** By integrating more advanced AI models and improving the NLP capabilities, the virtual counselor could offer more personalized and context-aware responses, helping users with a wider range of emotional needs.
2. **Improved Habit Tracker:** The habit tracker could be personalized with smarter recommendations and insights, allowing users to set and achieve goals based on their individual progress and behaviors.
3. **Expanded Book Recommendations:** Incorporating user ratings and reviews, as well as connecting with platforms like Goodreads, could make the book recommendation system more dynamic and tailored to specific self-care topics.
4. **Mobile App Development:** A mobile version of the website would increase accessibility, allowing users to track their habits, engage with the counselor, and receive book suggestions on-the-go.
5. **Professional Collaboration:** Integrating professional therapists for live sessions or workshops could offer users a more direct way to engage with qualified mental health support.
6. **Community Support:** Adding a community forum or peer support groups could encourage users to share experiences and advice, building a supportive environment for mental health.
7. **Multilingual Support:** Adding language options could make the platform accessible to a global audience, further expanding its reach.

These developments will help the platform evolve to meet the needs of users, ensuring it remains a valuable resource for self-care and mental health support.

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